Code No: 158DG **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year II Semester Examinations, September - 2022
FUNDAMENTALS OF ROBOTICS
(Common to CE, ME, ECE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the basic components of a robot system with the help of a line diagram.
b) Explain with principles of operation use of wrist force sensor with force sensing.[7+8]

2.a) Write a short note on internal and external sensors in Robotics.
b) Discuss the role of Robotics in Engineering. [8+7]

3.a) Write a short note on homogenous transformation in Robotics.
b) Differentiate magnetized gripper and pneumatic gripper. [7+8]

4.a) Explain the mechanical grippers and their linkage mechanisms with neat sketches.
b) Explain the concept of differential kinematics in Robotics. [8+7]

5.a) Differentiate between path planning and trajectory planning.
b) Determine the manipulator Jacobian matrix and singularities for the 3-DOF articulated arm. [7+8]

6.a) Write a short note on manipulator Jacobian in Robotics.
b) Solve the fifth order polynomial trajectory-planning problem with a suitable example.[7+8]

7.a) What are the different Robot programming languages?
b) Write the capabilities and limitations of Lead through programming. [7+8]

8.a) Sketch and explain different robot layout cells used in industrial applications. Discuss advantages, limitations and applications of each one.
b) What features are required for Robot in spray painting? [8+7]

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JNTUH Used Papers